ICT330

Examination - January Semester 2024

Database Management Systems

Monday, 15 April 2024 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **4** questions and comprises **7** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

ICT330 Copyright © 2024 Singapore University of Social Sciences (SUSS)

(Full marks: 100)

Question 1

The national sports hub opens some of its facilities for public booking from 7am to 10pm daily on an hourly basis. The rates (peak and non-peak rates) are dependent on the type of game. For example, the peak and non-peak rates for badminton are $8 and $4 respectively. The peak and non-peak rates for table tennis are $6 and $3 respectively. A person must have a sports account to make payment for booking. An account can be opened with an id, name, date of birth, home address and email.

To book a facility, a person must provide a game type, a datetime for an available slot and a facility number. Once a game is provided, the available datetimes for slots are shown with facility numbers of available facilities for those available slots. The location for each facility is indicated in a map. Facility numbers start from 1 in running order for each game type. For example, if there are 8 badminton courts in sports hub, the badminton courts are given running numbers 1 to 8, and if there are 12 tables for table tennis, they are given running numbers 1 to 12. A booking id is generated for each booking.

Before and after a game session, a staff of sport hub will make a quick check on the booked facility. Damages due to negligence are recorded to allow sports hub to recover cost of repairs/replacements. The itemised costs and descriptions are then emailed to the person who made the booking.

State your assumption(s) for only data requirements that are not specified.

Without adding any attribute/identifier not described in the question, construct a conceptual entity relationship (ER) model from the statements of requirements to represent the data model, showing the data requirements in **ONE** diagram.

Question 1a

Include entities, with entities name, relevant attributes and identifier:

(10 marks)

Question 1b

Include relationships, with maximum and minimum cardinalities and relationship name.

(10 marks)

Question 2

You are given the table:
ElderlyActivity(attendeeID, name, phone, activityDate, activityName, activityDescription, activityVenue)

This table is used by a centre to record the free activities it organises for the elderly in its community. These activities are held at fixed locations in order not to cause stress and confusion to the elderly, e.g., health talks are conducted in the conference room, screening tests are performed at a tentage at the backyard, and breakfast are held in the communal hall of the centre. It is necessary to record particulars of attendees to allow the centre to get fundings for the activities.

An activity may happen several times in a year or may happen regularly, for example, free breakfast is available every Tuesday and Friday. On some days, there may be two or more concurrent activities. You can assume that there will always some attendees for each organised activity. Assume also that attendees are identified by attendeeID and activities can be identified by activityName.

Question 2a

Formulate and list the functional and multi-valued dependencies (MVD). Do not introduce any new columns not in the given table. State your assumption(s) for only data requirements that are not specified.

(4 marks)

Question 2b

Draw dependency graph and propose candidate key(s) for the table. Show composite keys in brackets.

(4 marks)

Question 2c

(i) For each MVD in answer to Q2(a), explain why it is subsumed or not subsumed.

(ii) Normalise the table to Boyce Codd Normal Form (BCNF) and Fourth Normal Form (4NF) by applying the BCNF approach covered in the course text.

Show how you apply the steps to arrive at the tables, discuss and explain whether the structure of each derived table at each iteration is already in BCNF and 4NF (if applicable).

(10 marks)

Question 2d

Assemble your BCNF and 4NF tables and constraints clearly by giving each table and the referential integrity constraint(s), if any, in the forms described below:

Tables in this form:

TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key and
- indicate (FK) against the foreign keys, if any.

Referential integrity constraint(s), if any, **together with the child table** in this form:

TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in

TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(7 marks)

Question 3

Question 3a(i)

Transform the ERD in Figure Q3(a)(i) into a logical model.

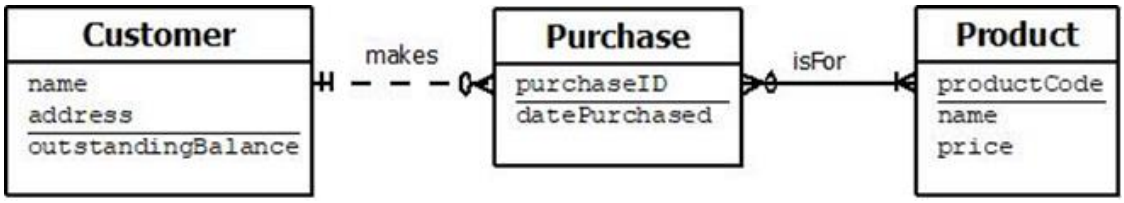

Figure Q3(a)(i)

Show the tables in the logical model in the form:

TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key,
- indicate (FK) against the foreign keys, if any, including integrity constraints in the form:

  TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in

  TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(6 marks)

Question 3a(ii)

Refer to the description of the columns for the column definition, and then code the appropriate SQL statements to implement **ONLY** the table **Purchase** in Figure Q3(a)(i). All columns are required.

name	At most 40 characters in length
address	At most 255 characters in length
outstandingBalance	A decimal number with 2 numbers after the decimal point. A value between 0 and 300
purchaseID	An integer, auto generated with starting value 1 and increment 1
datePurchase	Datetime with default datetime and must be not later than the current datetime
productCode	Consists of two alphabets followed by two digits.
product name	At most 40 characters in length
price	A decimal number with 2 numbers after the decimal point. A value between 5 and 50

Constraints for the relationships in Figure Q3(a)(i):

makes	delete and update are cascaded
isFor	delete disallowed and update disallowed

(6 marks)

Question 3b

For this question part, you are allowed to use inline views in your query formulation, but you must NOT create views using the CREATE VIEW statement.

Use the database tables in Figure Q3b.

	ArtPieceCode	title
1	500	Memories IV
2	511	Surf and Bird
3	521	The Tilled Field
4	522	La Lecon de Ski
5	523	On White II
6	524	Woman with a Hat
7	537	The Woven World
8	548	Night Bird
9	551	Der Blaue Reiter
10	552	Angelus Novus
11	553	The Dance
12	554	I and the Village
13	555	Claude Monet Painting
14	561	Sunflower
15	562	The Fiddler
16	563	Spanish Dancer
17	564	Farmer's Market #2
18	565	Farmer's Market #2

ArtPiece table

	CustomerID	lastName	FirstName	City	Country
1	1000	Janes	Jeffrey	Renton	USA
2	1001	Smith	David	Loveland	USA
3	1015	Twilight	Tiffany	Langley	USA
4	1033	Smathers	Fred	Bainbridge Island	USA
5	1034	Frederic...	Mary Beth	Denver	USA
6	1036	Warning	Selma	Vancouver	Canada
7	1037	Wu	Susan	Atlanta	USA
8	1040	Gray	Donald	Bodega Bay	USA
9	1041	Johnson	Lynda	Washington	USA
10	1051	Wilkens	Chris	Olympia	USA
11	1052	Tan	Peter	Vancouver	Canada

Customer table

	TransactionID	AcquisitionPrice	AskingPrice	SalesPrice	CustomerID	ArtPieceCo
1	100	30000.00	45000.00	42500.00	1000	500
2	101	250.00	500.00	500.00	1015	511
3	102	125.00	250.00	200.00	1001	521
4	103	250.00	500.00	400.00	1034	522
5	104	250.00	250.00	200.00	1001	523
6	105	200.00	500.00	400.00	1034	524
7	115	1500.00	3000.00	2750.00	1033	537
8	121	15000.00	30000.00	27500.00	1015	548
9	125	125.00	250.00	200.00	1001	551
10	126	200.00	400.00	NULL	NULL	552
11	127	125.00	500.00	400.00	1034	553
12	128	125.00	250.00	225.00	1036	554
13	129	125.00	250.00	225.00	1036	555
14	151	10000.00	20000.00	17500.00	1036	561
15	152	125.00	250.00	225.00	1001	562
16	153	200.00	400.00	350.00	1001	563
17	154	250.00	500.00	400.00	1040	564
18	155	250.00	500.00	NULL	NULL	565

Transaction table

Note that row 10 and 18, the art pieces have been purchased by the gallery but are not sold to customers yet. For the other rows where customerID is indicated, the art pieces have been sold off.

Figure Q3b

Formulate SQL queries to perform the following tasks:

(i) List the customers in USA who live in cities starting with the letter L. List in descending alphabetical order of lastName.

	CustomerID	lastName	FirstName	City	Country
1	1015	Twilight	Tiffany	Langley	USA
2	1001	Smith	David	Loveland	USA

(ii) List the art pieces that have been purchased by customers in Canada. Use subquery without any join.

	ArtPieceCode	title
1	554	I and the Village
2	555	Claude Monet Painting
3	561	Sunflower

(iii) List the customer and the number of purchases made as well as their purchase. Include customers without purchases.

	CustomerID	lastName	FirstName	NumPurchases	ValueOfPurchases
1	1000	Janes	Jeffrey	1	42500.00
2	1001	Smith	David	5	1175.00
3	1015	Twilight	Tiffany	2	28000.00
4	1033	Smathers	Fred	1	2750.00
5	1034	Frederickson	Mary Beth	3	1200.00
6	1036	Warning	Selma	3	17950.00
7	1037	Wu	Susan	0	0.00
8	1040	Gray	Donald	1	400.00
9	1041	Johnson	Lynda	0	0.00
10	1051	Wilkens	Chris	0	0.00
11	1052	Tan	Peter	0	0.00

.

(15 marks)

Question 3c

Write a trigger to ensure that sales price must not be less than 50% of the acquisition price when a customer makes an art piece purchase:
State any assumptions you made in implementing the trigger.

(8 marks)

Question 4

Question 4a

Demonstrate the basic techniques of concurrency control and their application.

Consider the facility booking scenario in Q1. Suppose two or more persons are booking the same game and the same facility (e.g., same badminton court) for the same slot.
Discuss a problem that can arise when a wrong isolation level is used. You should state the isolation level that leads to the problem discussed as well as to explain how the problem can arise.
Then propose an isolation level where the problem will not happen. Explain why the proposed isolation level cannot cause the problem to happen.

(6 marks)

Question 4b

Describe **ONE (1)** permission that should be given to customers using the database for Question 3, based on the ERD shown in Figure Q3(a)(i). Describe **ONE (1)** permission that should not be given to customers. Specify the permissions in terms of these actions: select, delete, insert and update on user tables only.

(4 marks)

Question 4c

Given Figure 4(c), explain what the database recovery procedure will do for the transactions T1 to T3. Explain also how the transaction log help ensure the ACID properties of T1 to T3.

Record Pointer	Transaction ID	Reverse Pointer	Forward Pointer	Time	Type of Operation	Table	Row ID	Before Image	After Image
1	T1	0	2	11:01:00	Start				
2	T1	1	3	11:01:36	Delete	Table1	2	2, ...	
3	T1	2	8	11:02:55	Delete	Table2	14	14, 2, ...	
4	T2	0	5	11:03:01	Start				
5	T2	4	9	11:03:24	Update	Table3	15	15, 3, ...	15,5, ...
6	T3	0	7	11:04:41	Start				
7	T3	6	9	11:05:05	Delete	Table1	3	2, ...	
8	T1	3	0	11:05:42	Rollback				
Checkpoint									
9	T2	4	0	11:25:00	Rollback				
10	T5	7	11	11:27:25	Insert	Table1	14		14, 8, ...
11	T3	10	0	11:28:33	Commit				
Crash!									

Figure 4(c)

(10 marks)

----- END OF PAPER -----

ICT330 Copyright © 2024 Singapore University of Social Sciences (SUSS)